Wazuh SIEM Project – Evidence Report

18 October 2026

Project Overview

- Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard services are running.
- Log sources: Windows 10 (Agent ID 001, 192.168.254.128) and Kali Linux (Agent ID 002, 192.168.254.131); both are Active.
- Windows and Kali events are visible in Threat Hunting → Events.
- The wazuh-alerts-retention-7d policy deletes indices after 7 days, and wazuh-alerts-retention-30d deletes indices after 30 days. In the October 18, 2026 screenshot, the 7-day policy is attached to indices 2026.10.17 and 2026.10.18, while the 30-day policy is attached to indices 2026.10.01, 2026.10.04, and 2026.10.05.
- This report documents five tested security detection use cases, with corresponding Wazuh alerts and rule/configuration evidence.

Action Plan

Project 4: "SIEM Detection Engineering, Alert Triage & SOC Operations"

WEEK	FOCUS	OWNER	STATUS
Week 1	SIEM Architecture & Log Onboarding	AA Ahmed Adel	✓ Done
Week 2	Correlation Rules & Detection Use Cases	AA Ahmed Adel	✓ Done
Week 3	Dashboards, Metrics & Alert Triage	AS Amir Sobhy	Pending
Week 4	Threat Intel, SOAR & AI-Assisted SOC Workflow	MA Mohammed Ahmed	Pending

SIEM Architecture Diagram

The diagram shows the confirmed integrated log sources: Windows Security and Sysmon logs from Windows 10, and Authentication (PAM) and system logs from Kali Linux, flowing to the Wazuh Manager for analysis and onward indexing and visualization.

Retention policy status as of October 18, 2026: the 7-day policy is attached to indices wazuh-alerts-4.x-2026.10.17 and wazuh-alerts-4.x-2026.10.18; the 30-day policy is attached to indices wazuh-alerts-4.x-2026.10.01, wazuh-alerts-4.x-2026.10.04, and wazuh-alerts-4.x-2026.10.05.

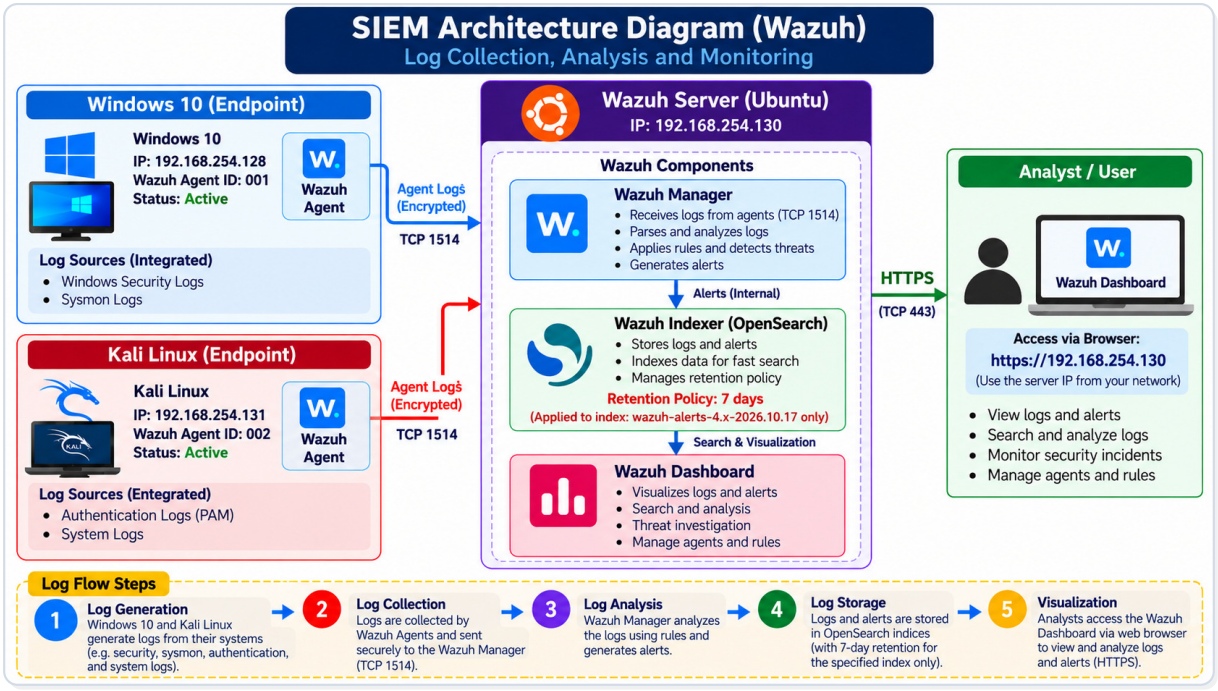
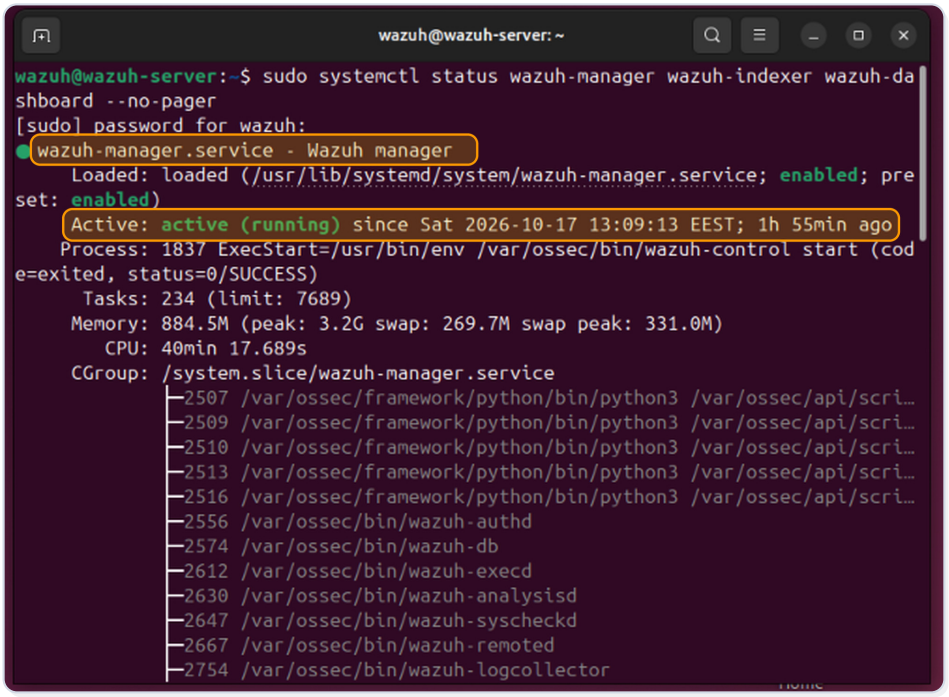


Figure: Wazuh SIEM lab architecture.

Screenshot Evidence

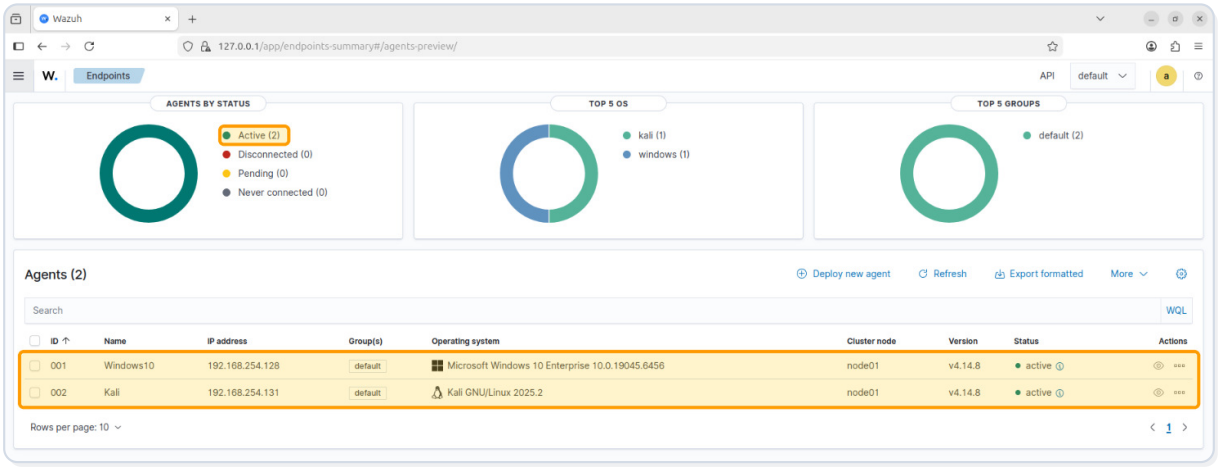
01. Core Wazuh Services

The terminal output shows the Wazuh services in a running state.



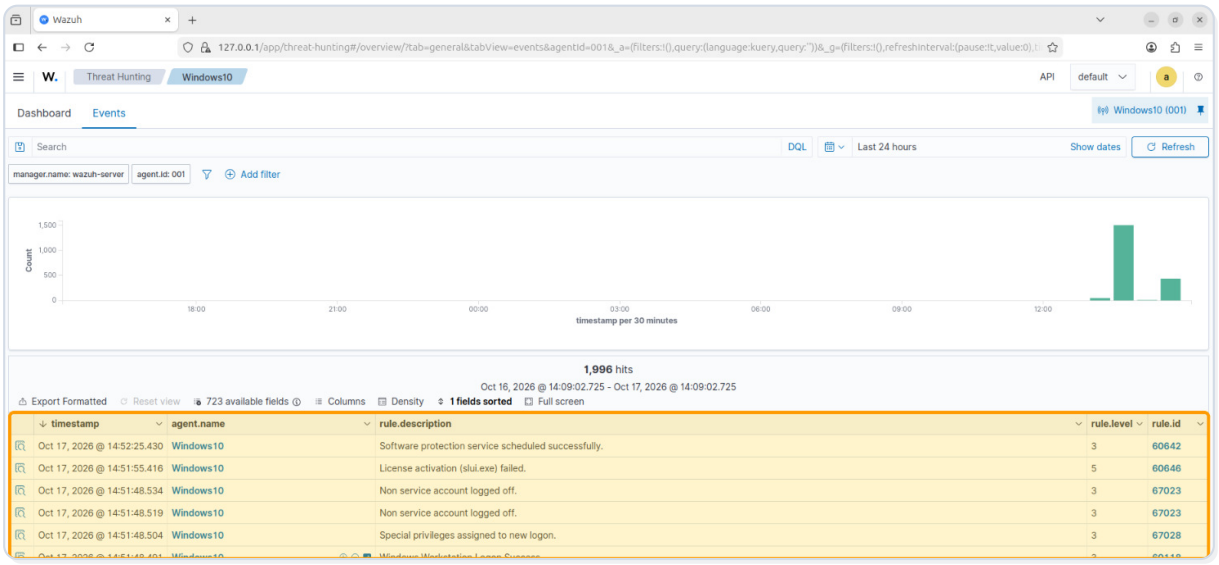
02. Connected Log Sources

The Endpoints dashboard lists Windows10 (ID 001) and Kali (ID 002), both Active.



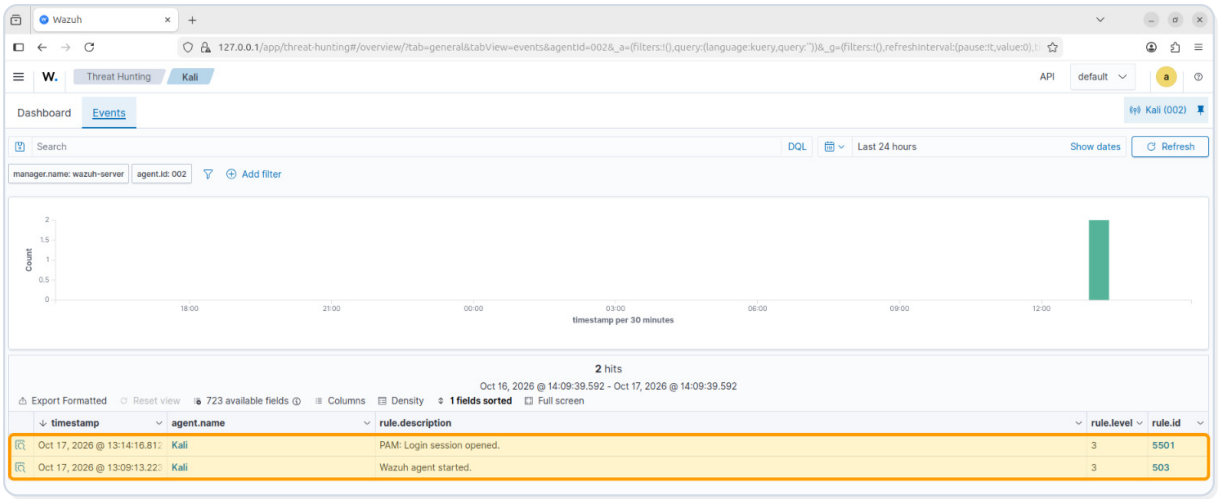
03. Windows Events in Wazuh

Threat Hunting → Events displays Windows10 events with rule IDs and severity levels.



04. Kali Events in Wazuh

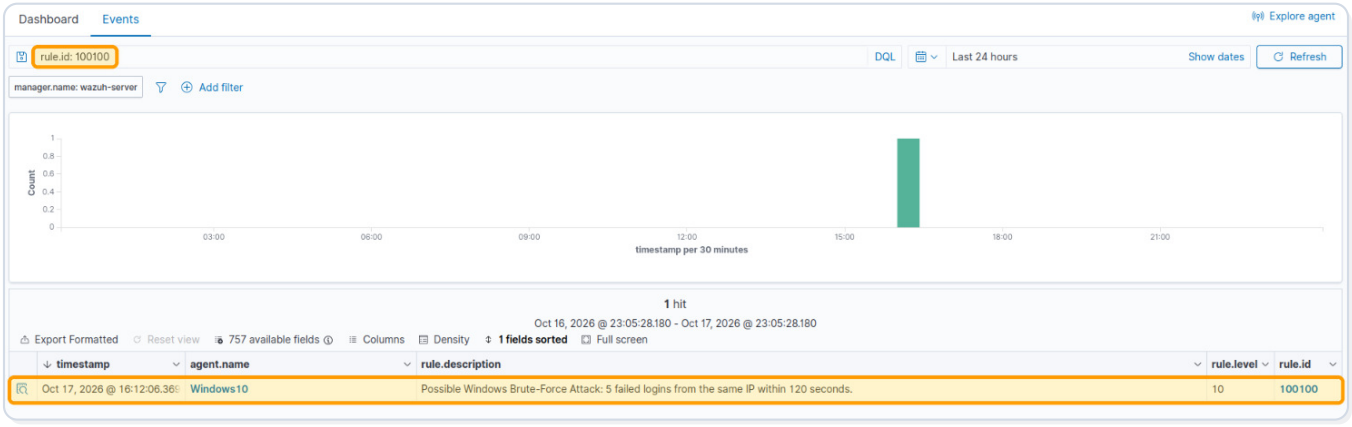
Threat Hunting → Events displays the PAM login-session and Wazuh agent-started events from Kali.



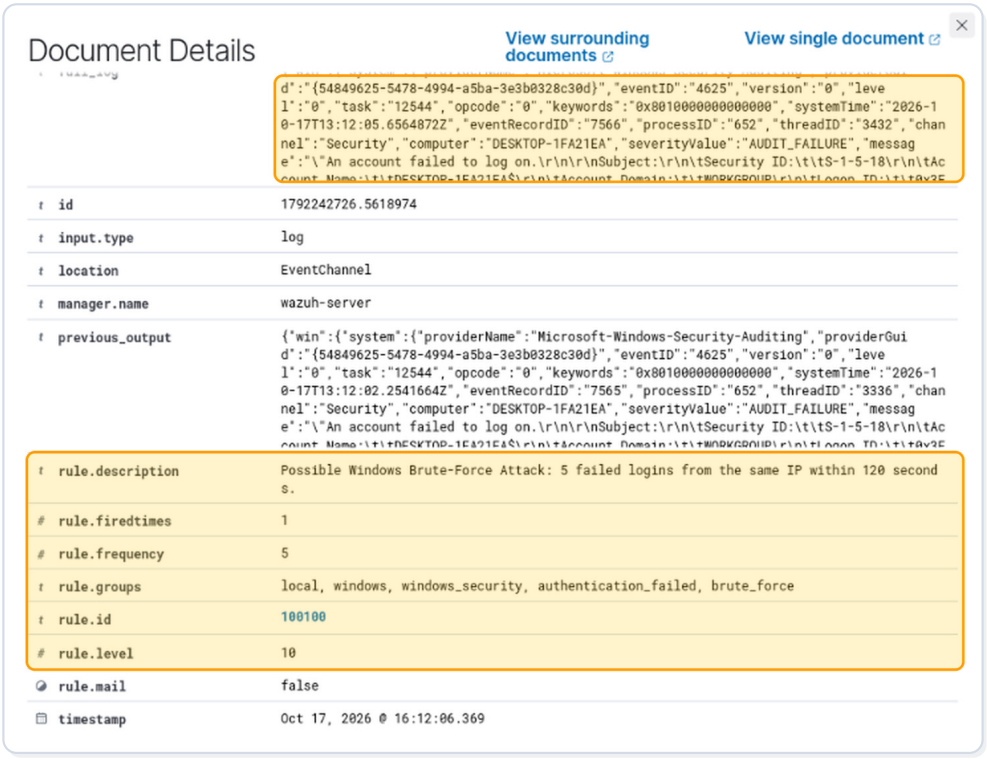
Use Case 1 — Brute-Force Attack Detection

The following evidence is attached for Use Case 1: the Wazuh alert, event details, JSON view, and custom detection rule definition.

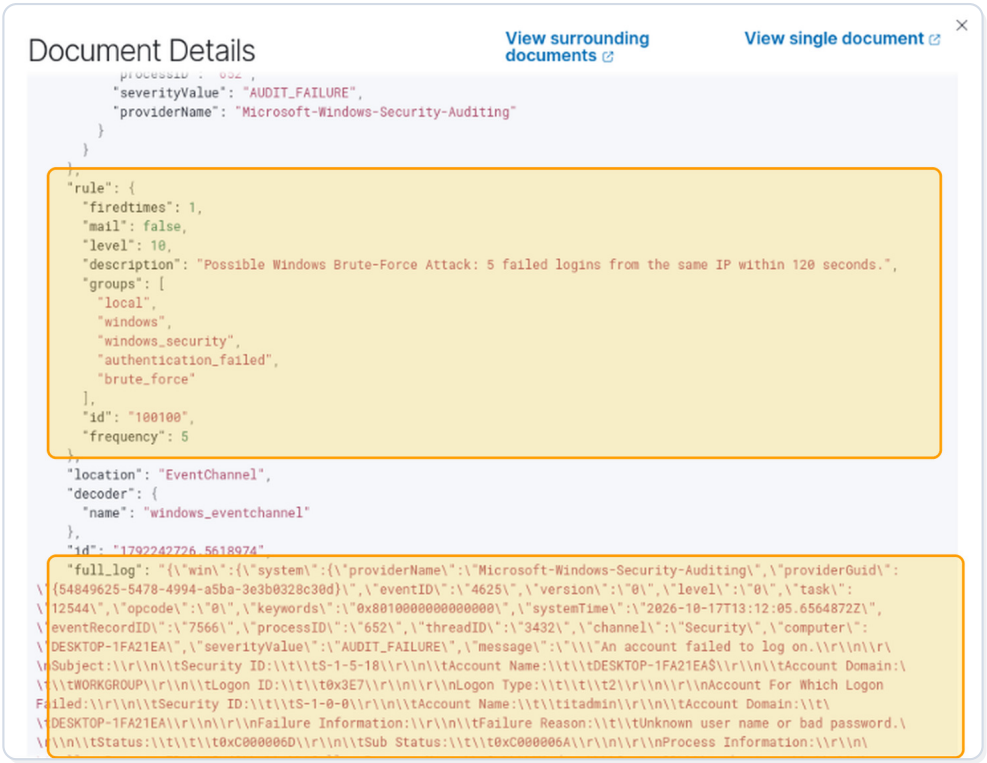
Evidence 1: Brute-Force alert in the Wazuh Dashboard showing Rule ID 100100 and Level 10.



Evidence 2: Document details showing Event ID 4625, the alert description, Rule ID, and level.



Evidence 3: JSON view of the alert data and detection rule properties.



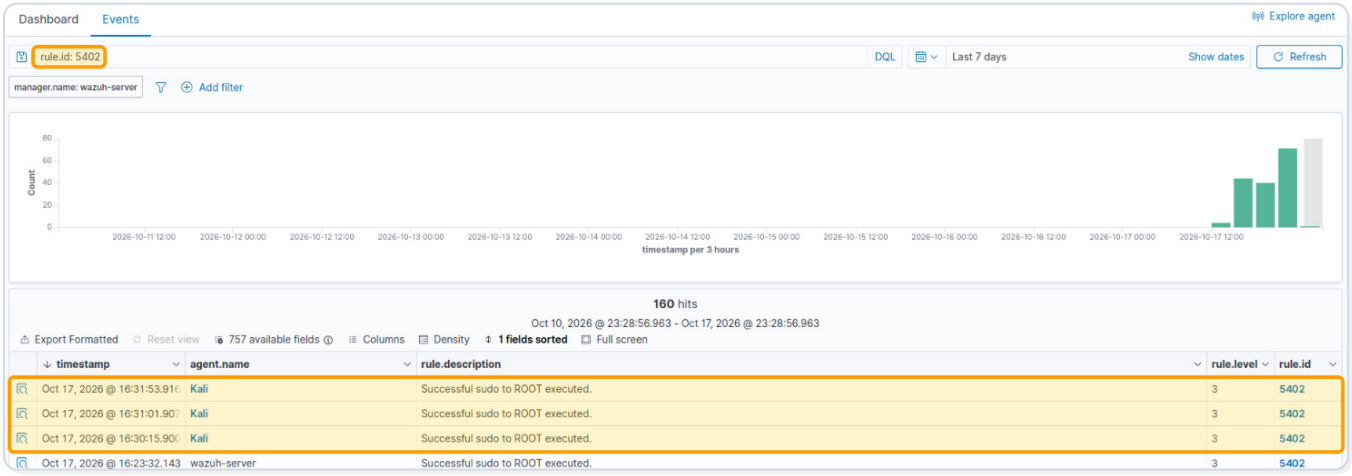
Evidence 4: Rule 100100 definition in local_rules.xml, including frequency=5, timeframe=120, and same-source-IP matching.

```
wazuh@wazuh-server:~$ sudo grep -n -B 5 -A 20 'id="100100"' /var/ossec/etc/rules/local_rules.xml
[sudo] password for wazuh:
19-</group>
20-
21-
22-<group name="local,windows,windows_security,">
23-
24:  <rule id="100100" level="10" frequency="5" timeframe="120">
25-    <if_matched_sid>60122</if_matched_sid>
26-    <same_field>win.eventdata.ipAddress</same_field>
27-    <description>Possible Windows Brute-Force Attack: 5 failed logins from the same IP within 120 seconds.</description>
28-    <group>authentication_failed,brute_force,</group>
29-  </rule>
```

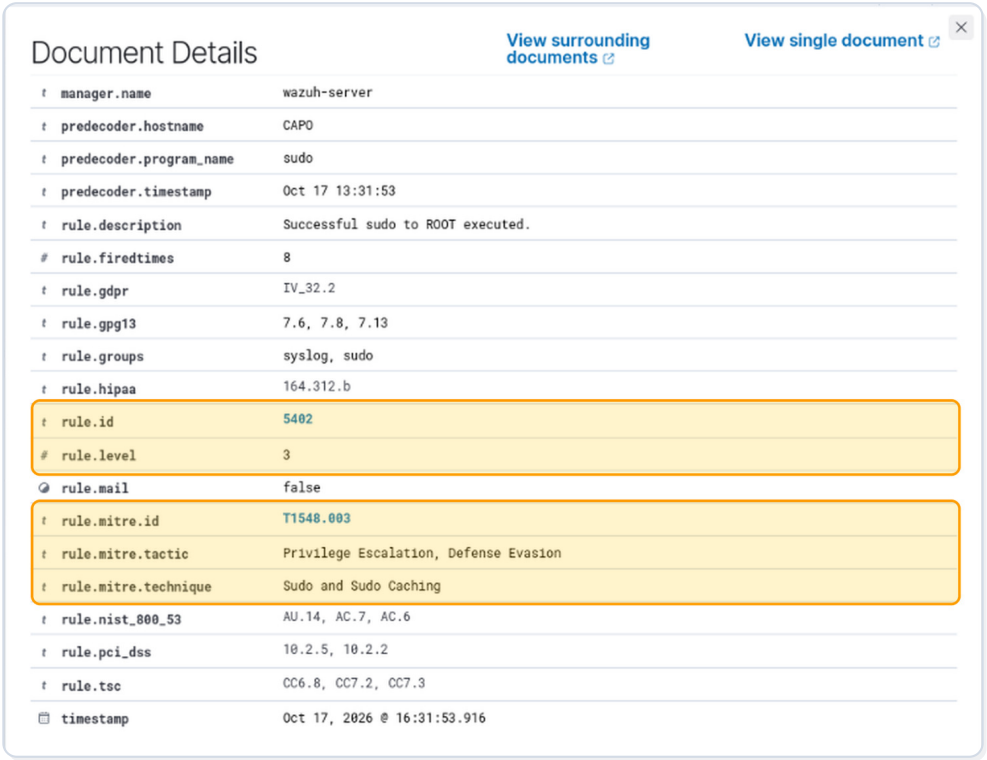
Use Case 2 — Suspicious Sudo / Privilege Escalation

Evidence for Use Case 2: Wazuh sudo alerts, event details, the recorded command, and confirmation of root access on Kali Linux. Rule 5402 is a native level-3 rule, not a custom high-severity rule.

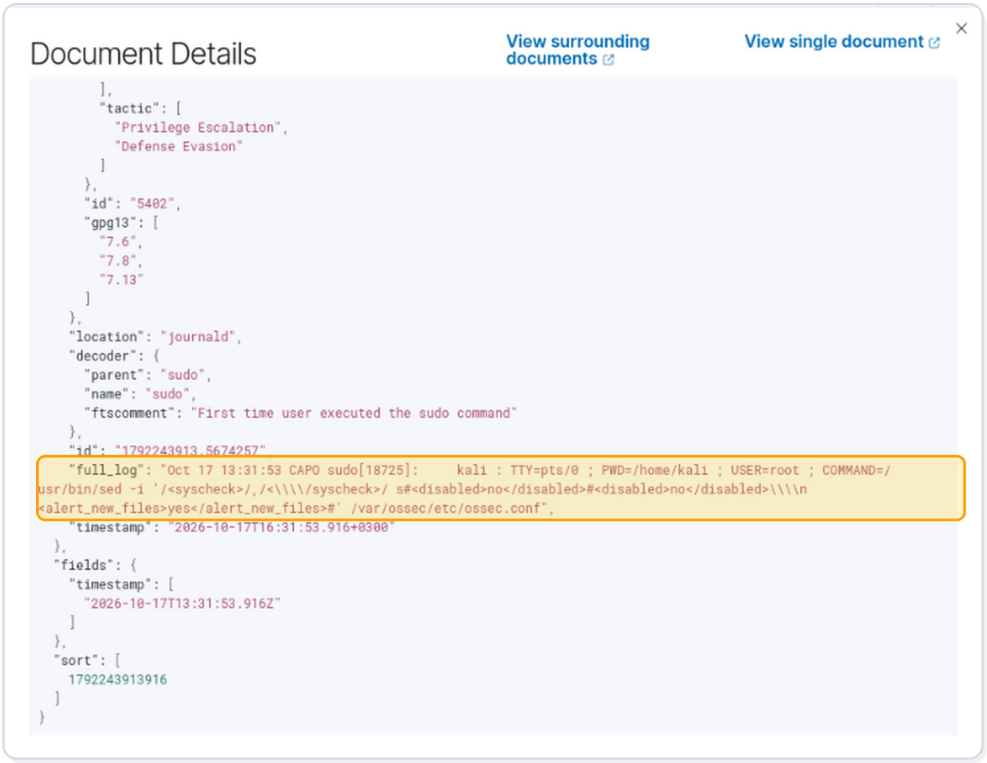
Evidence 1: Successful sudo to ROOT events in the Wazuh Dashboard for the Kali agent, Rule ID 5402 and Level 3.



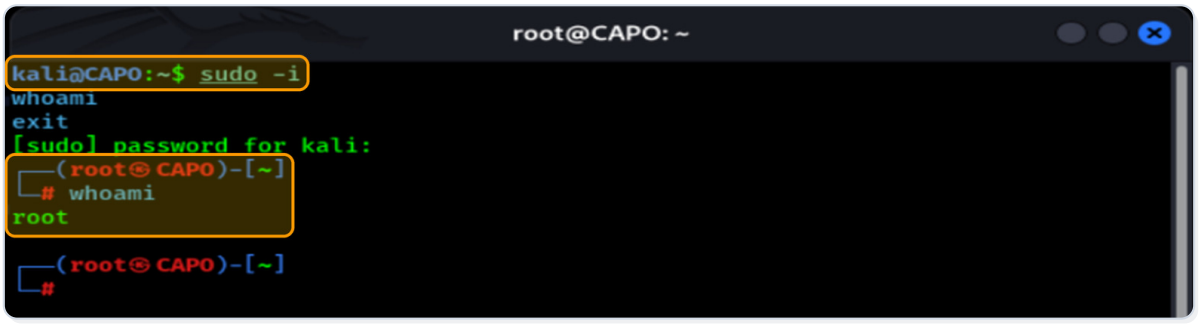
Evidence 2: Sudo alert details, including Rule ID 5402, Level 3, and MITRE ATT&CK information.



Evidence 3: JSON view showing the command recorded in the sudo event's full_log.



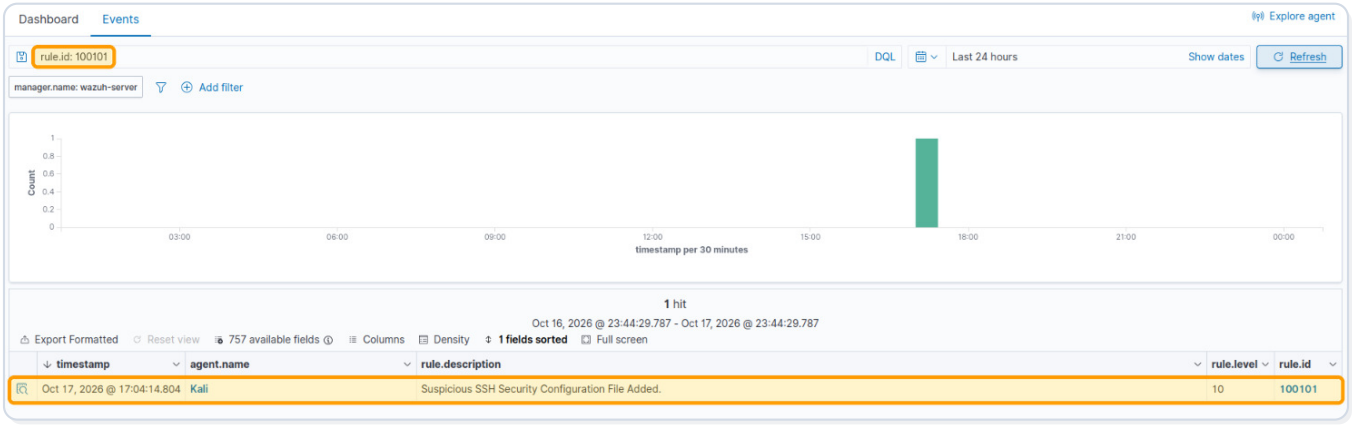
Evidence 4: Kali Linux terminal showing whoami returning root after sudo -i.



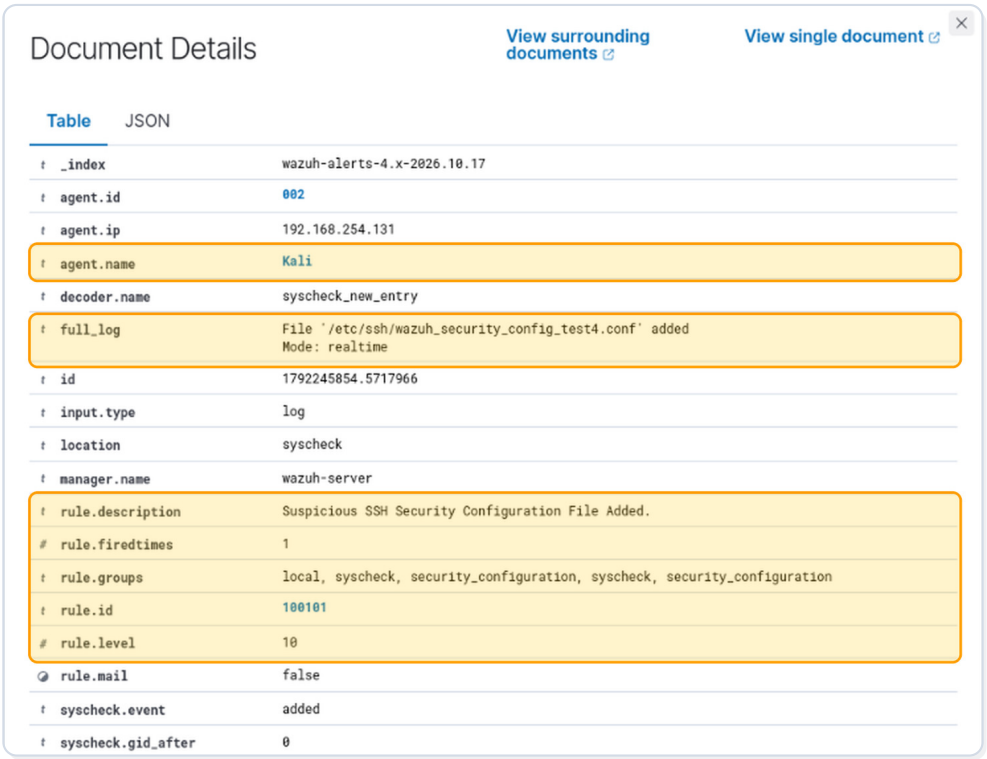
Use Case 3 — File Integrity Monitoring (FIM)

The evidence shows custom Rule ID 100101 at level 10 on the Kali agent when a file was added under /etc/ssh, recorded as a syscheck added event.

Evidence 1: Wazuh Dashboard result for Rule ID 100101, showing the custom detection rule at level 10.



Evidence 2: Alert details showing the Kali agent, file path /etc/ssh/wazuh_security_config_test4.conf, and rule description, ID, and level.



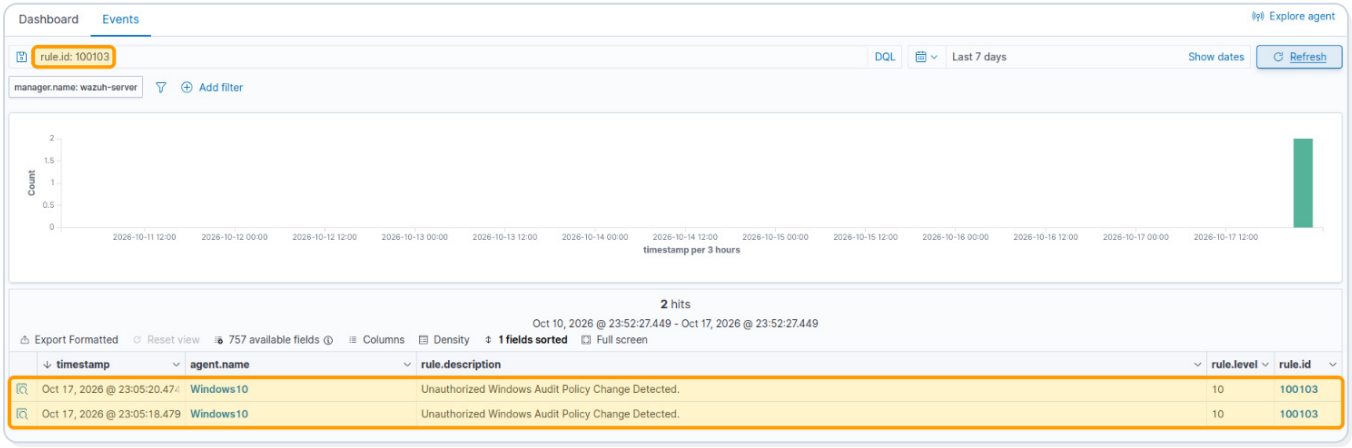
Evidence 3: JSON view confirming Rule ID 100101 and the file-addition event recorded by syscheck.



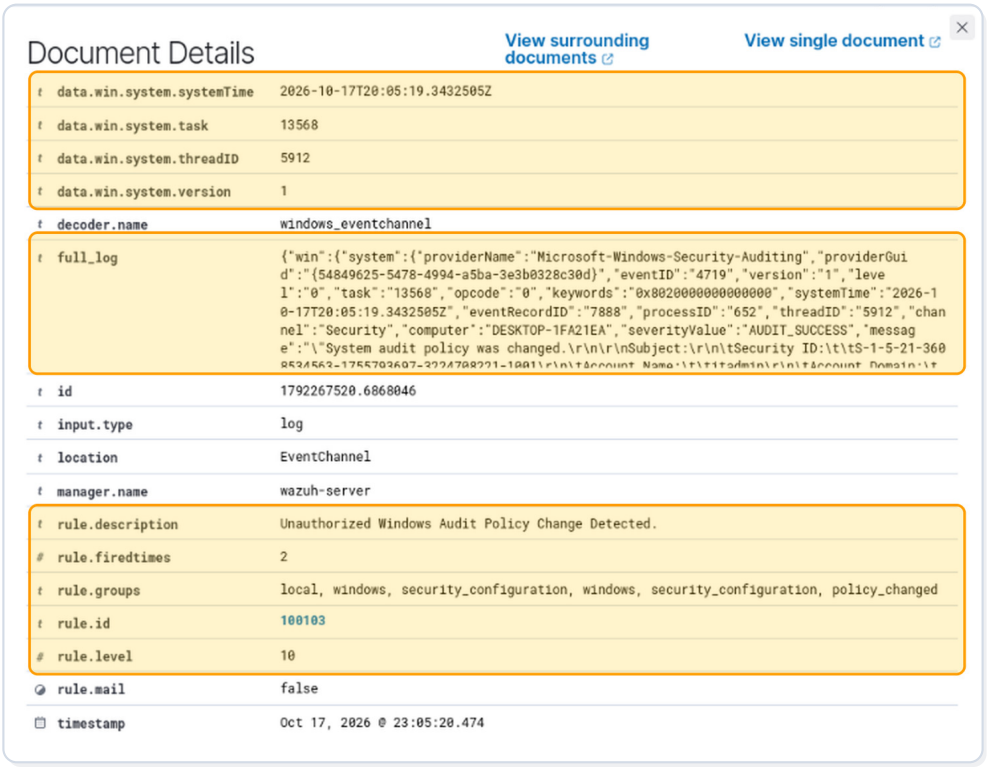
Use Case 4 — Windows Audit Policy Change Detection

The evidence shows custom Rule ID 100103 at level 10 on Windows10, matched to Windows Security Event ID 4719 for an Audit Policy change. Two matching alerts are shown.

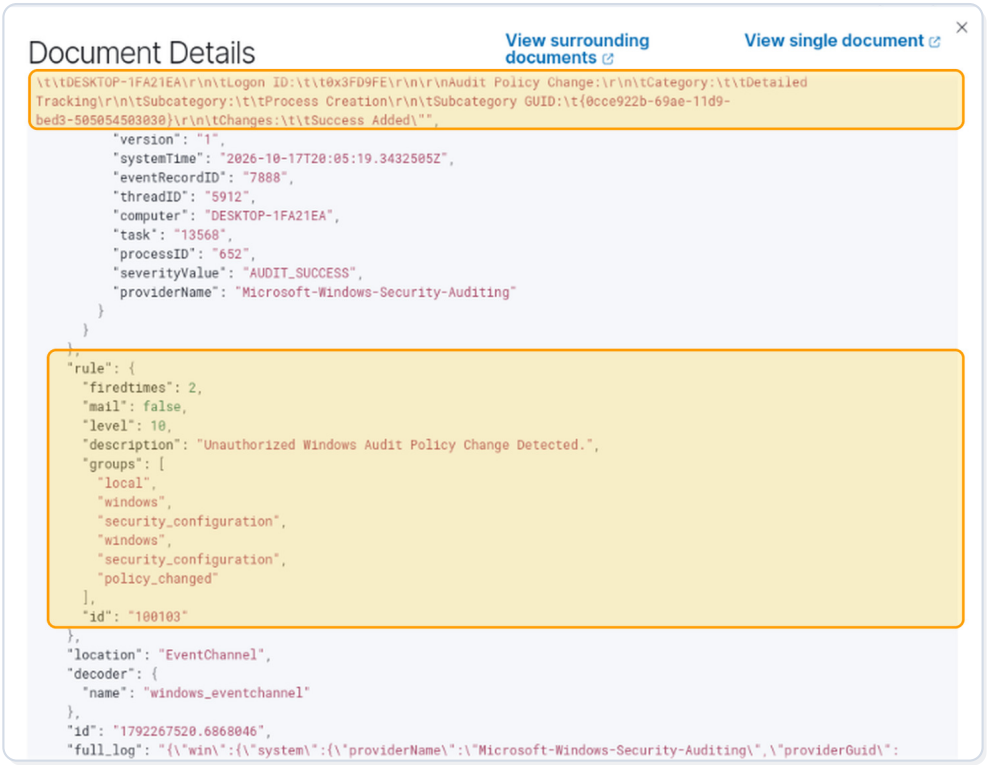
Evidence 1: Wazuh Dashboard results for Rule ID 100103, showing two level-10 alerts.



Evidence 2: Event details showing Windows10, Windows Security Event ID 4719 data, and the rule description, ID, and level.



Evidence 3: JSON view showing the Audit Policy change and the custom alert metadata.

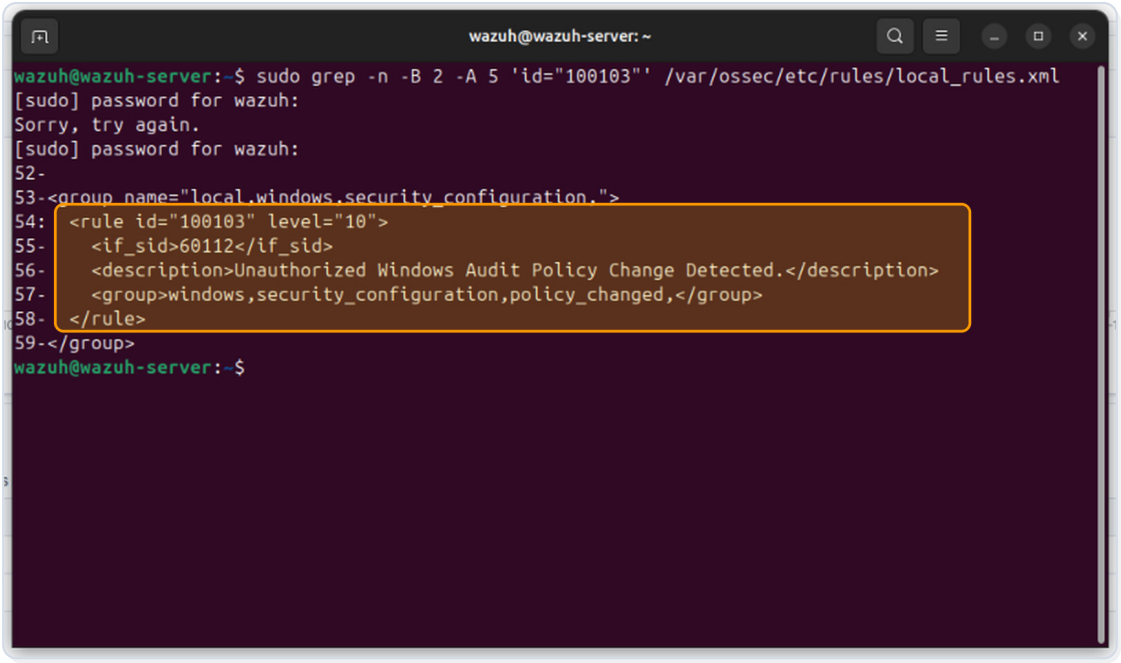


Additional Evidence — Use Case 4

The following evidence documents the Windows Audit Policy detection rule and the related Windows audit configuration.

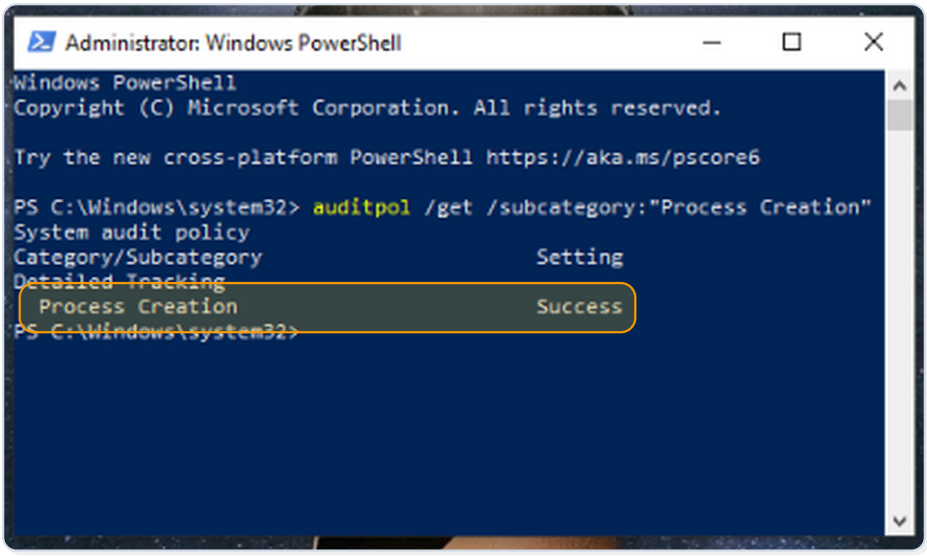
Additional Evidence 4: Windows Audit Policy Rule Definition

Shows Rule 100103 at level 10, its parent rule 60112, and the alert description.



Additional Evidence 5: Windows Audit Policy Verification

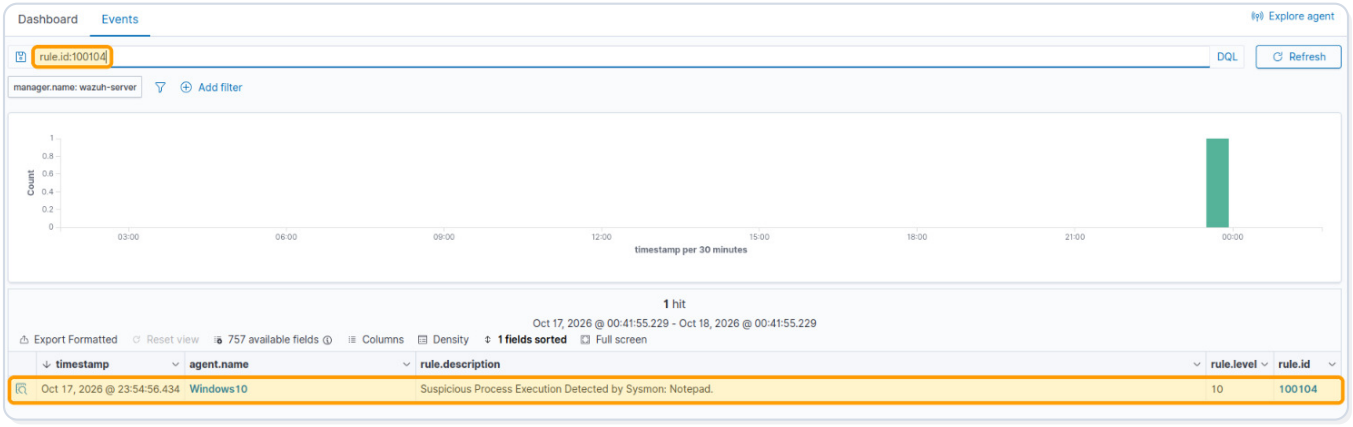
Shows Process Creation auditing configured for Success at the time of capture.



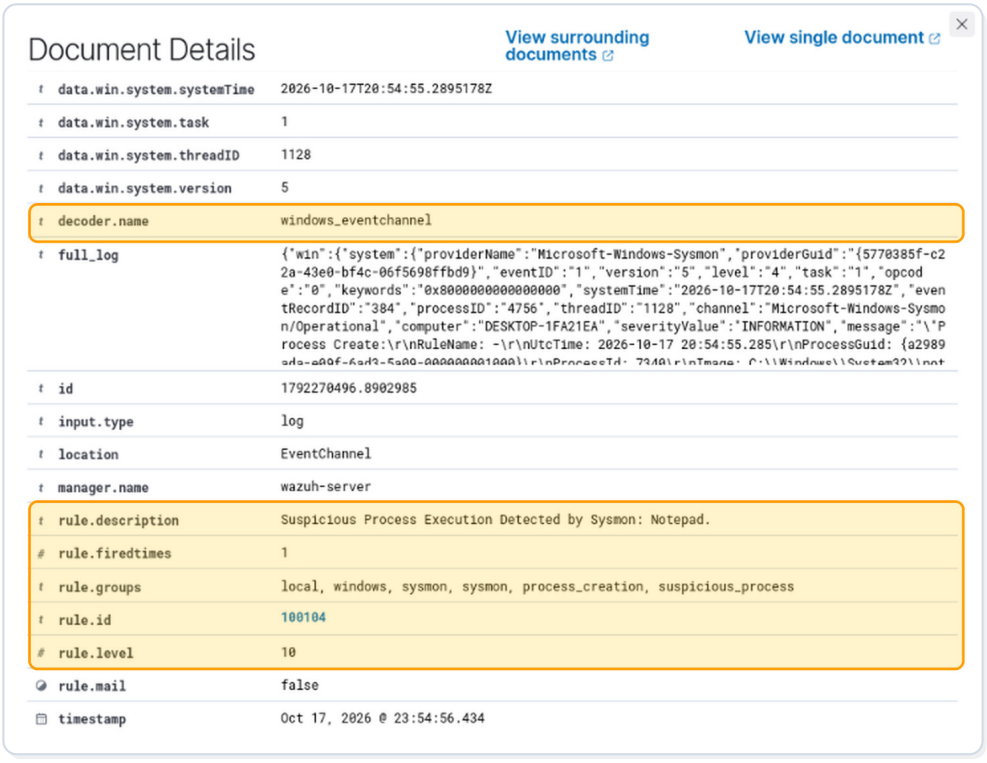
Use Case 5 — Suspicious Process Execution (Sysmon Event ID 1)

The evidence confirms that Sysmon Event ID 1 (Process Creation) reached Wazuh from the Windows10 agent and triggered custom Rule ID 100104 at level 10 when notepad.exe was executed. The alert includes process details, command line, and parent process.

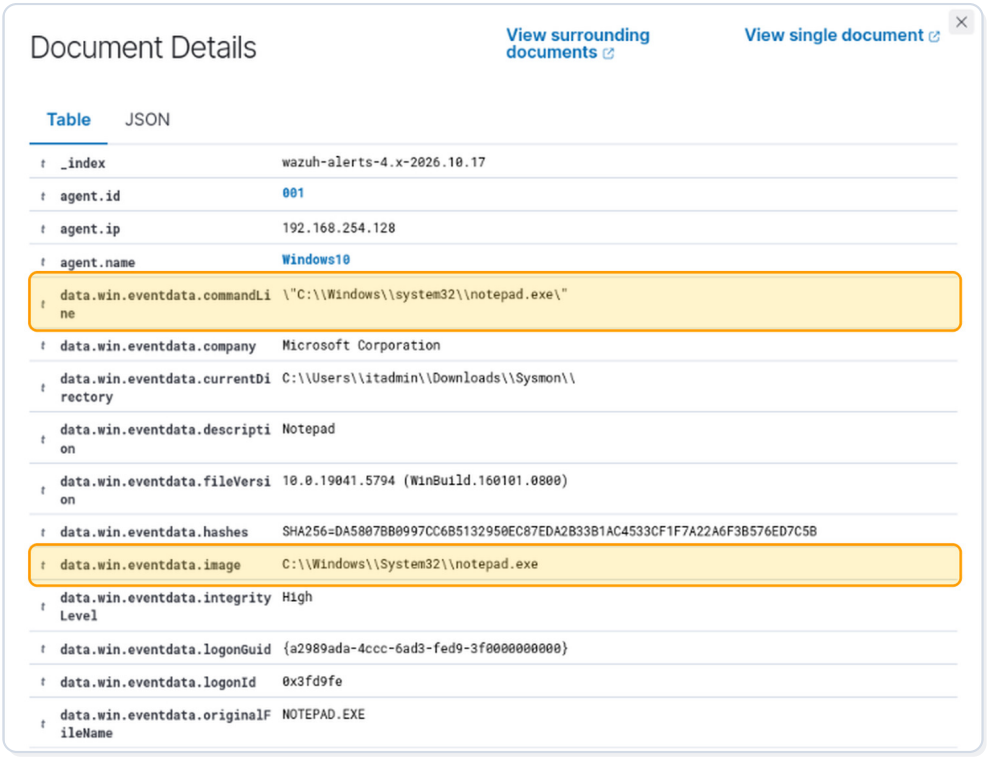
Evidence 1: Wazuh Threat Hunting result filtered by Rule ID 100104, showing one level-10 alert on Windows10.



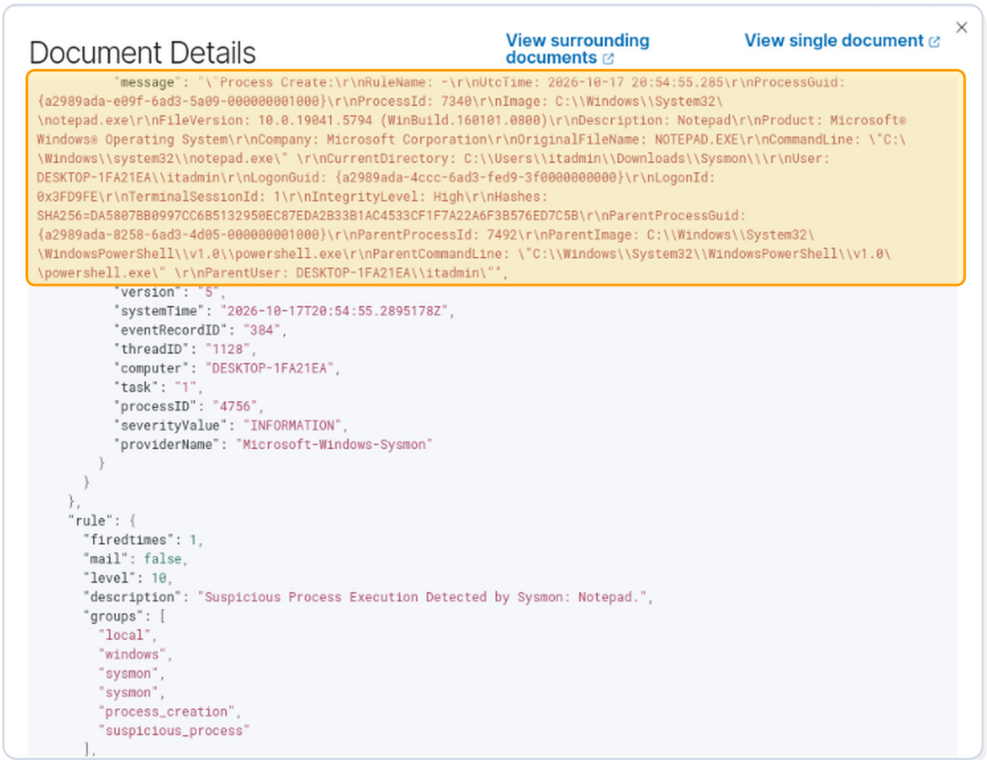
Evidence 2: Alert details showing Rule ID 100104, level 10, Windows10 agent, and Windows Event Channel decoder.



Evidence 3: Event fields showing Event ID 1 process creation and the notepad.exe image path.



Evidence 4: JSON details showing Sysmon Event ID 1, Notepad process creation, command line, and parent process.

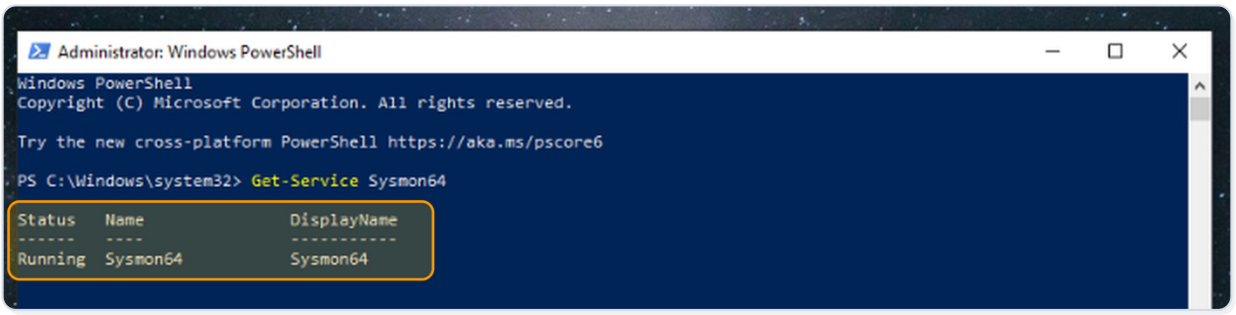


Additional Evidence — Use Case 5

The following evidence documents the Sysmon service, Wazuh Agent collection configuration, and delivery of Sysmon logs to the Wazuh server.

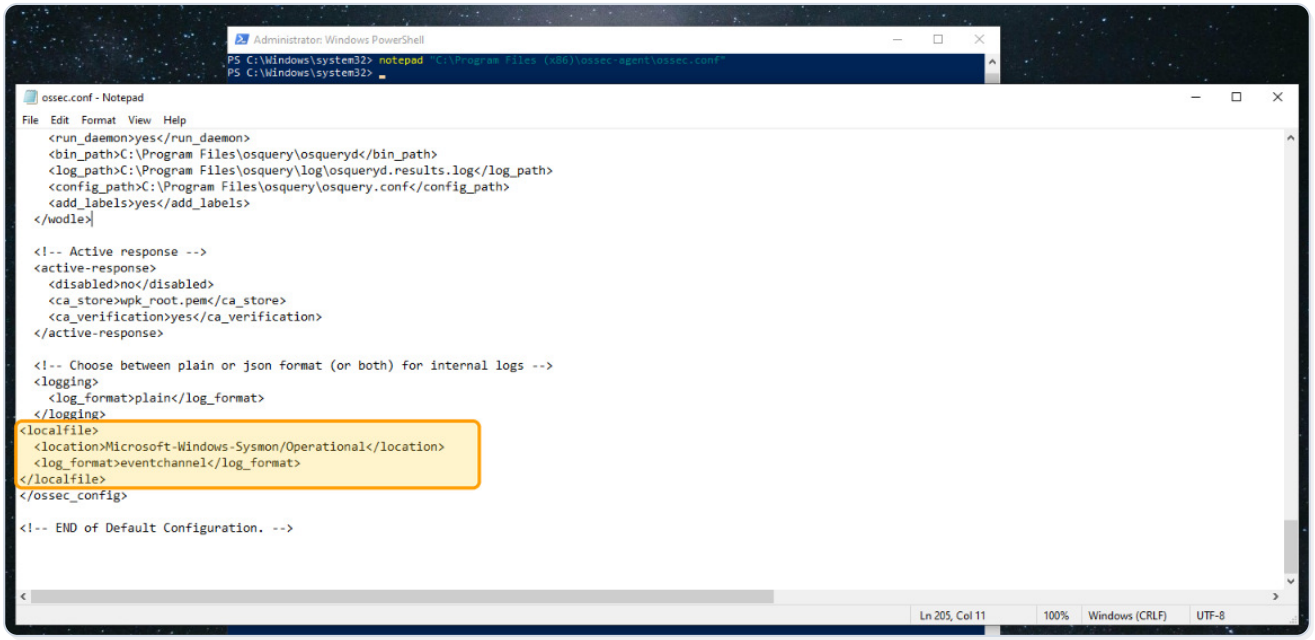
Additional Evidence 5.1: Sysmon Service Status

The screenshot shows the Sysmon64 service running on Windows.



Additional Evidence 5.2: Wazuh Agent Sysmon Collection Configuration

The ossec.conf screenshot shows a localfile entry collecting the Microsoft-Windows-Sysmon/Operational channel using the eventchannel log format.



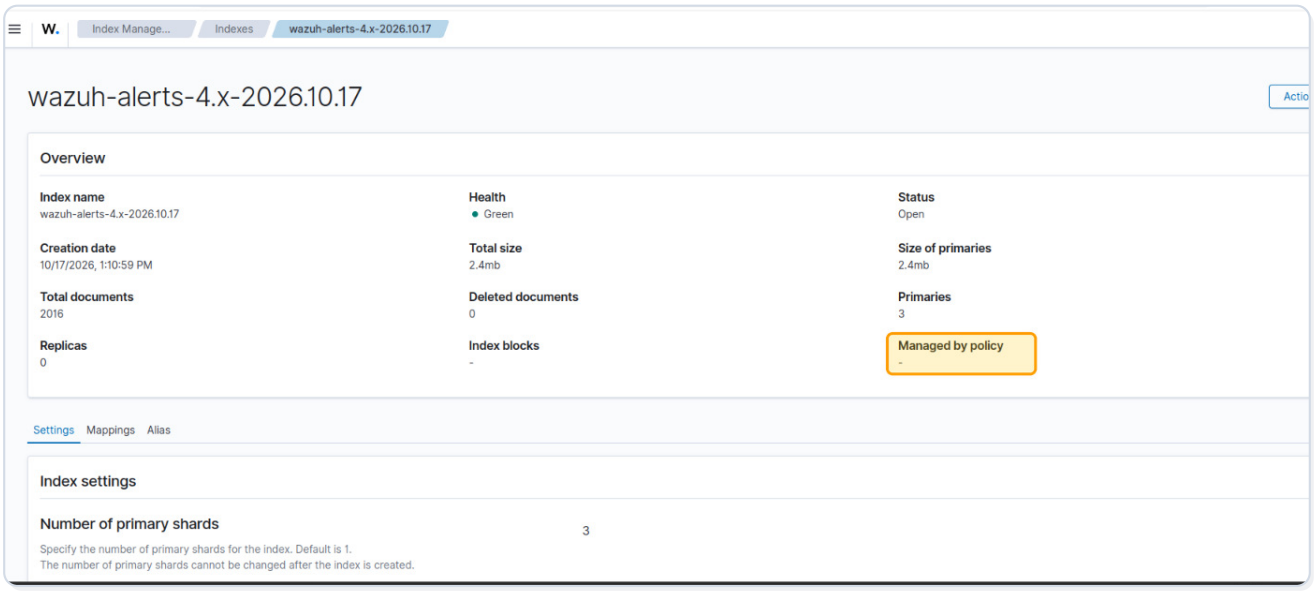
Additional Evidence 5.3: Sysmon Event ID 1 Received by Wazuh Server

The archives.json output shows Sysmon logs received from the Windows10 agent (ID 001), including Event ID 1 for creation of notepad.exe, alongside the level-10 custom alert from Rule ID 100104.



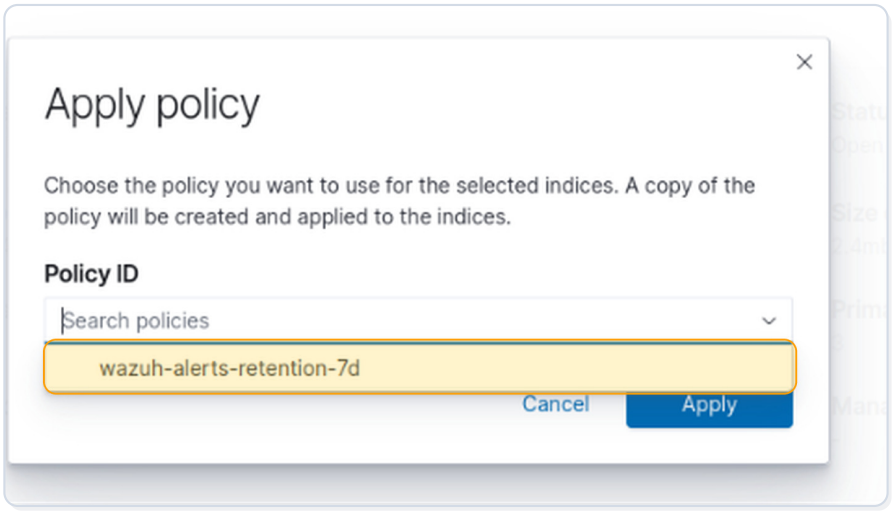
05. Index Details Before Policy Application

The index details before applying the policy do not show a managed-by-policy value.



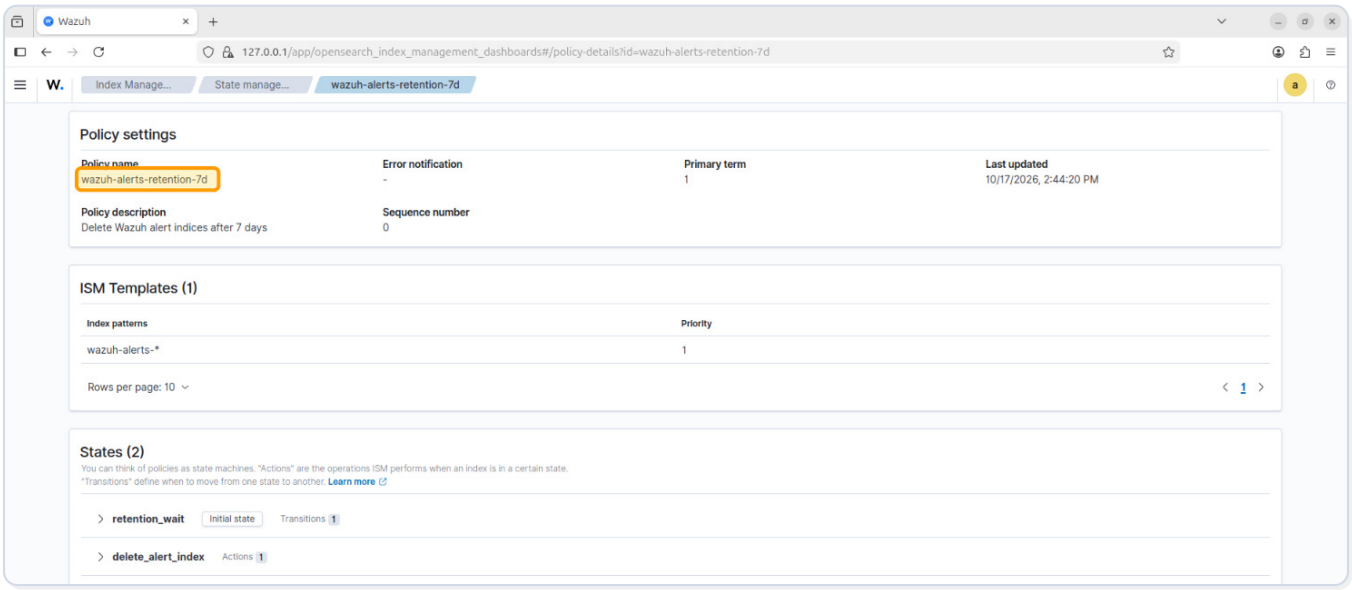
06. Retention Policy Selection Dialog

The Apply policy dialog shows wazuh-alerts-retention-7d selected for application.



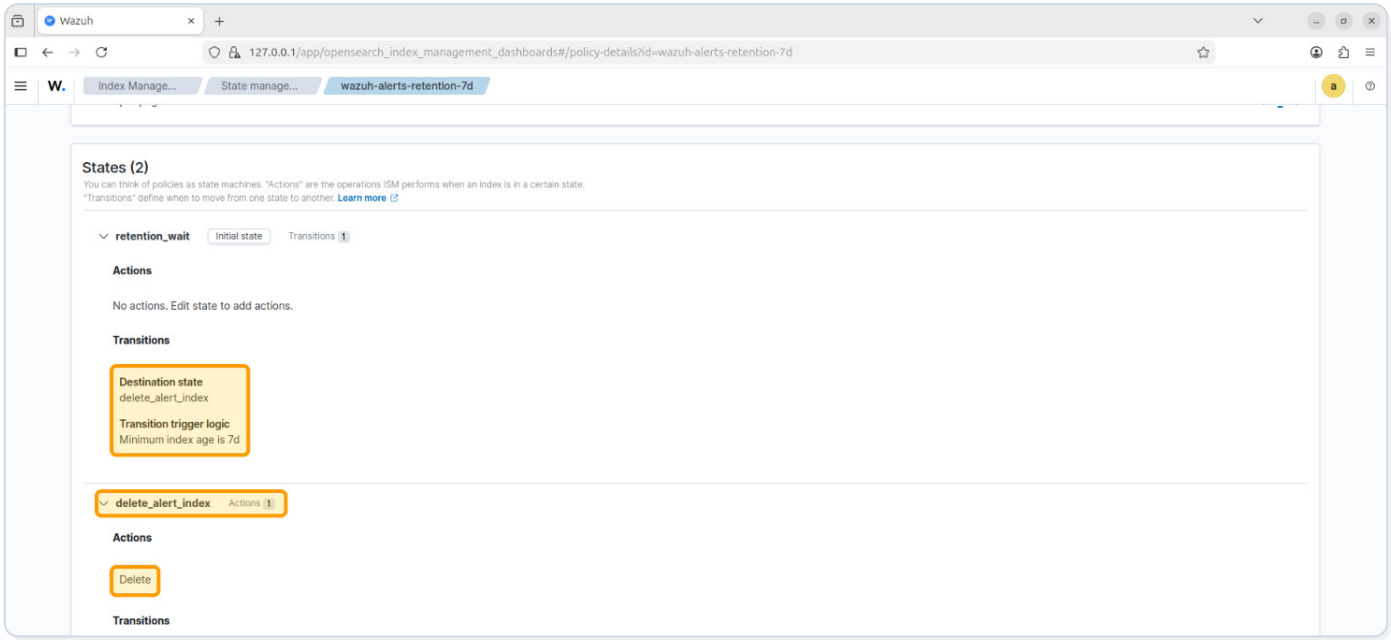
07. Policy Applied to an Index

The index details show Managed by policy: wazuh-alerts-retention-7d.



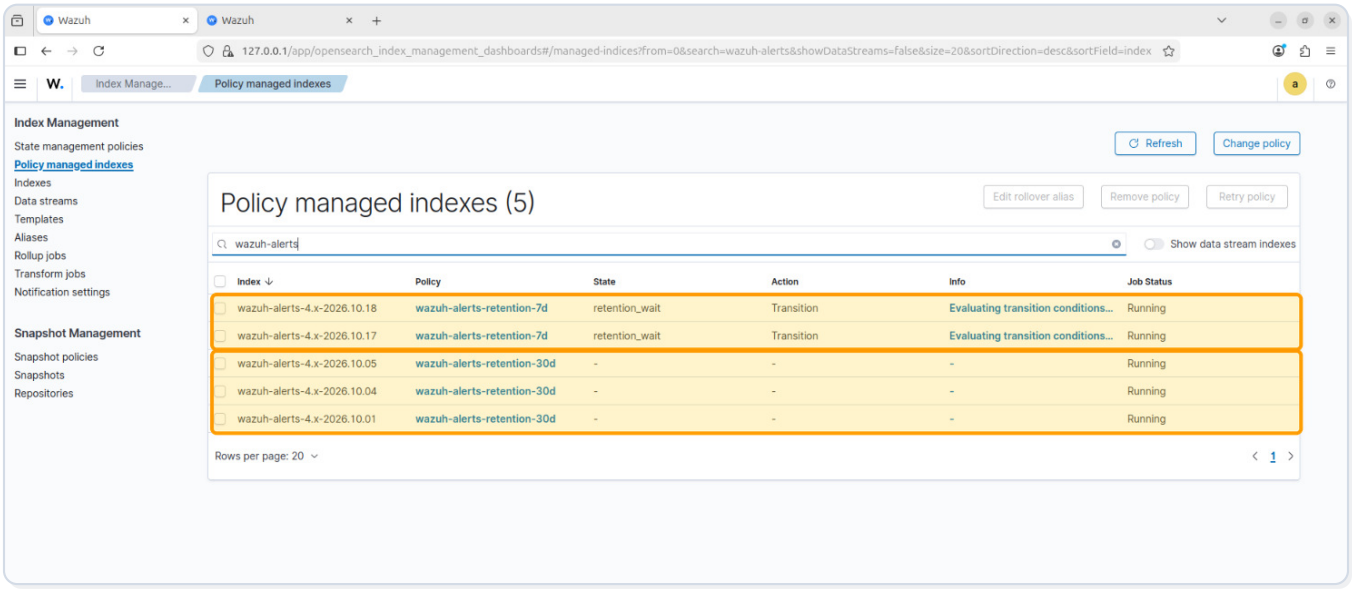
08. Retention Policy Details

The policy transitions when minimum index age reaches 7 days to delete_alert_index, which contains the Delete action.



09. Evidence of Retention Policies Applied to Current Indices

The screenshot shows wazuh-alerts-retention-7d attached to indices 2026.10.17 and 2026.10.18, and wazuh-alerts-retention-30d attached to indices 2026.10.05, 2026.10.04, and 2026.10.01. Job Status is Running; this confirms policy attachment, not that deletion has already occurred.



Evidence screenshot from the Policy managed indices page, October 18, 2026.